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End –End Communication at Transport Layer

a) Implement echo client server using TCP/UDP sockets.
b) Implement a chat program using socket programming.

AIM: To implement (a) an echo client-server using TCP and UDP sockets, and (b) a chat program using socket programming in Python.

tcp_server.py

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.bind(("127.0.0.1", 5000))

server.listen(1)

print("TCP Echo Server started...")

conn, addr = server.accept()

print("Connected:", addr)


while True:

   data = conn.recv(1024)

   if not data:

      break

   print("Client:", data.decode())

   conn.send(data)


conn.close()

server.close()
```

tcp_client.py

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("127.0.0.1", 5000))

print("Connected to TCP Echo Server")

while True:
    msg = input("Enter message: ")
    if msg.lower() == "exit":
        break

    client.send(msg.encode())
    data = client.recv(1024)

    print("Server:", data.decode())

client.close()
```

udp_server.py

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server.bind(("127.0.0.1", 5001))

print("UDP Echo Server started...")

while True:
    data, addr = server.recvfrom(1024)
    print("Client:", data.decode())

    if data.decode().lower() == "exit":
        break

    server.sendto(data, addr)

server.close()
```

udp_client.py

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

print("Connected to UDP Echo Server")

while True:
    msg = input("Enter message: ")

    client.sendto(msg.encode(), ("127.0.0.1", 5001))

    if msg.lower() == "exit":
        break

    data, addr = client.recvfrom(1024)
    print("Server:", data.decode())

client.close()
```

chat_server.py

```
import socket
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 5002))
server.listen(1)
print("Chat Server started...")
conn, addr = server.accept()
print("Connected:", addr)
while True:
    msg = conn.recv(1024).decode()

    if msg.lower() == "exit":
        break

    print("Client:", msg)

    reply = input("You: ")
    conn.send(reply.encode())

    if reply.lower() == "exit":
        break

conn.close()
server.close()
```

chat_client.py

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("127.0.0.1", 5002))

print("Connected to Chat Server")

while True:
    msg = input("You: ")
    client.send(msg.encode())

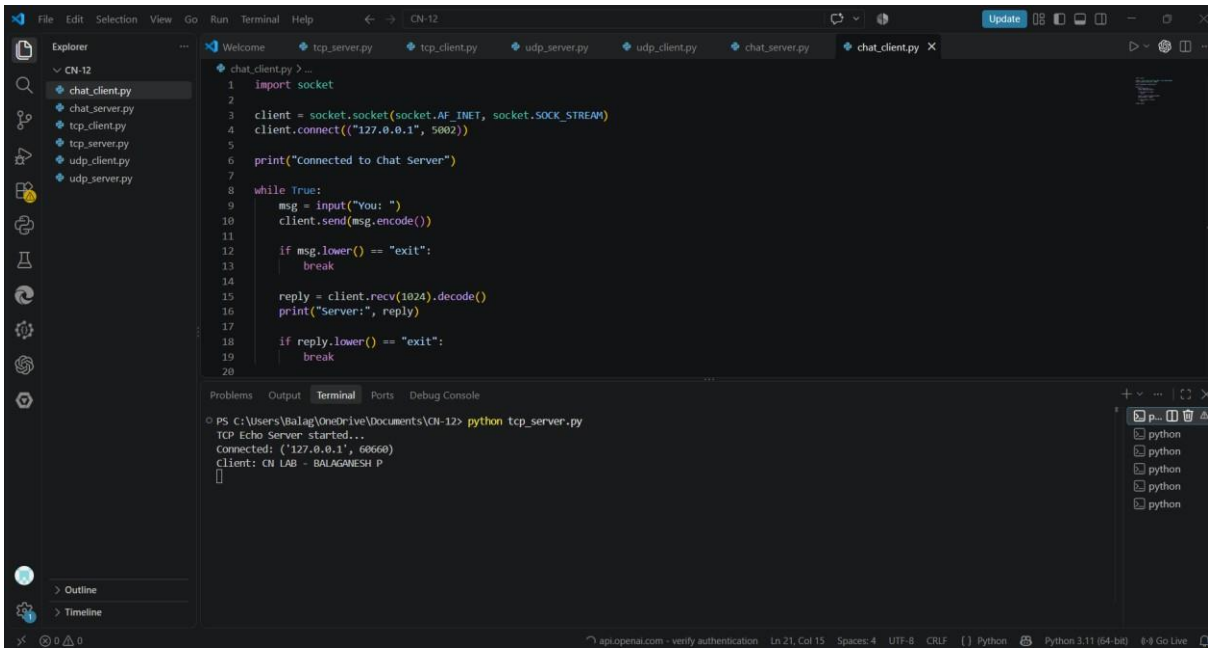
    if msg.lower() == "exit":
        break

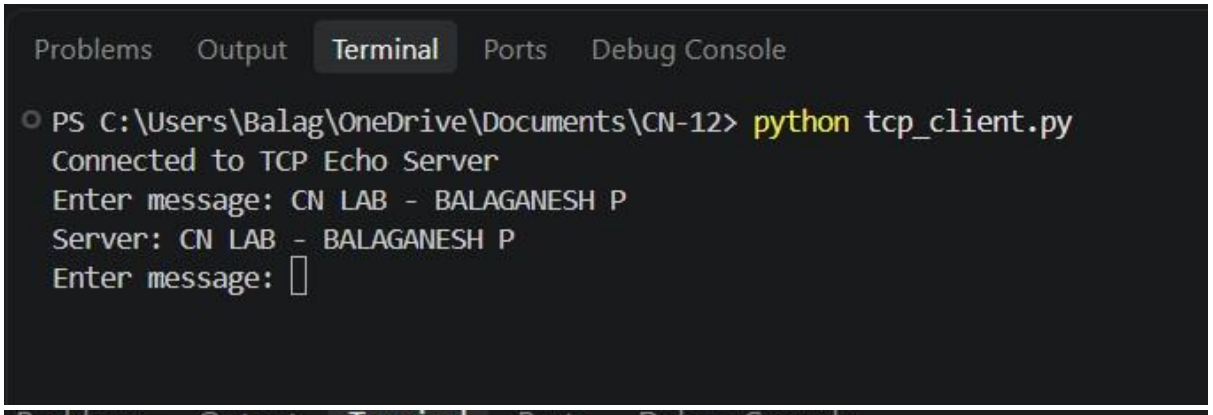
    reply = client.recv(1024).decode()
    print("Server:", reply)

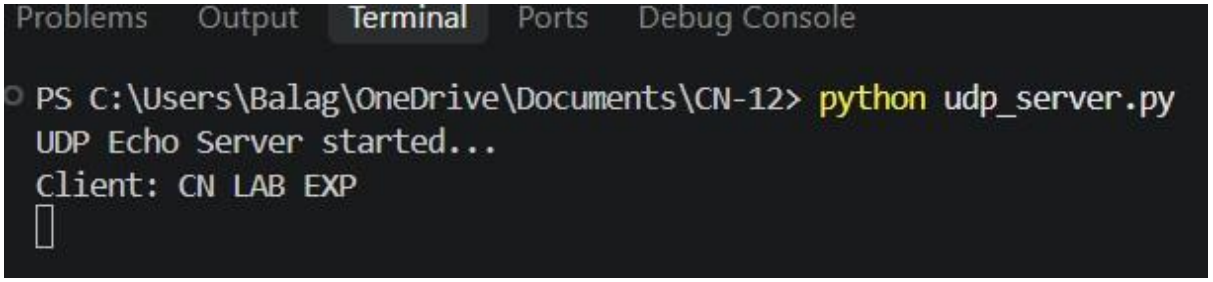
    if reply.lower() == "exit":
        break

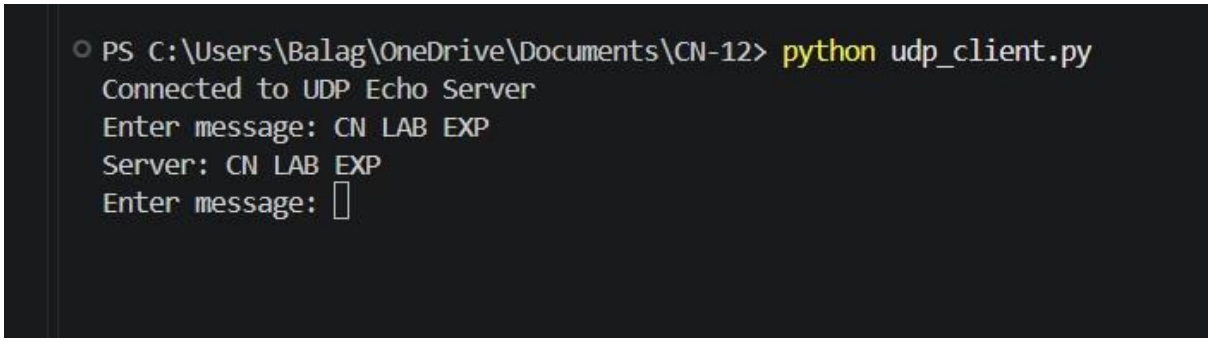
client.close()
```

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```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python chat_server.py
Chat Server started...
Connected: ('127.0.0.1', 58511)
Client: HII
You: GOOD MORNING
Client: WELCOME
You:
```

```
Problems  Output  Terminal  Ports  Debug Console

PS C:\Users\Balag\OneDrive\Documents\CN-12> python chat_client.py
Connected to Chat Server
You: HII
Server: GOOD MORNING
You: WELCOME
```